Decision Log

monday.com Business Intelligence Agent — Skylark Drones

Live prototype: [`https://skylark-bi-agent-aoqx2me7kbvpoapptdjtpr.streamlit.app/`] **Source:** [`https://github.com/Deepika240406/skylark-bi-agent.git`] **Stack:** Python · monday.com GraphQL API · pandas · Google Gemini (function calling) · Streamlit

1. Key assumptions

- **"Energy sector" does not exist in the data.** The brief's own example question asks about the energy sector, but the boards contain Mining, Renewables, Railways, Powerline, Construction and Others. The agent resolves Energy to **Renewables + Powerline**, states that mapping in its answer, and proceeds. Returning zero results would have been the worse failure.
- **The boards join on deal name, not client code.** Work Orders uses `WOCOMPANY_002`; Deals uses `COMPANY089`. There is *zero* literal overlap — two independently masked namespaces. The numeric parts appear to align if the `WO` prefix is stripped, but joining on that coincidence would attribute revenue to the wrong client. Deal name matches ~90% of work orders; the unmatched remainder is reported, not hidden.
- **Missing means unknown, never zero.** Totals are computed over records that have a value and reported with explicit coverage. Nothing is extrapolated from an average. A founder mistaking an estimate for an actual is precisely the failure this system exists to prevent.
- **"This quarter" means the current calendar quarter,** stated inline rather than asked about. In practice most date-filtered questions return nothing, because `Close Date (A)` is blank on 318 of 344 deals — so the agent falls back to the unfiltered view and explains why.
- **A recurring contract past its nominal end date is not late.** Seven work orders carry the status "Executed until current month" — monthly contracts running as designed. Including them put "on schedule" at the top of the delay list; separating them cut the overdue count from 49 to 42 and surfaced the genuinely stuck projects underneath.
- **Quantities are never summed across units.** `Quantities as per PO` mixes hectares, acres, route-kilometres, days and months. The cleaner splits number from unit, so cross-unit aggregation is impossible by construction.
- **'Tender' is a deal type, not an industry.** It appears in the sector column and is relabelled `Tender (not a sector)` so it can be excluded rather than silently counted as one.

2. Trade-offs chosen, and why

- **Direct GraphQL over MCP.** Fewer moving parts, full control over pagination and retries, faster to debug in six hours. MCP is a heavier integration for no gain on a read-only workload.
- **Fixed analysis functions over LLM-generated code.** The model chooses among seven functions and writes prose around the returned dict — it never calculates. Fabricated figures are therefore structurally impossible: no number can reach an answer except through `analytics.py`. The cost is a fixed question surface; anything outside those shapes gets an honest "I can't answer that."
- **Fetch-and-cache over per-question queries.** Both boards load once per session and are held in memory; querying on every message would be slow and rate-limited. A Refresh button clears the cache and re-queries live. **No CSV is read anywhere** — there is no file I/O in the codebase at all, and stripping the credentials makes the app produce nothing.
- **Business data is never hardcoded; domain vocabulary is, deliberately.** Something must know that "Won" means won and "Executed until current month" describes a healthy contract; no algorithm infers that. Those meanings are declared once at the top of `analytics.py` and are environment-overridable (`STATUS_WON`, `STATUS_ON_SCHEDULE`, `SMALL_SAMPLE_THRESHOLD`), so a different monday account needs a config change, not a code change.
- **Streamlit over a React front end.** A separate SPA plus API would have consumed two of the six hours in deployment plumbing and risked the one required deliverable. Streamlit was given a custom design system instead — cartographic palette, monospace figures, and a ticked coverage rule per column as the signature element.
- **Gemini on a free tier.** No billing dependency for a hosted demo. The trade-off is a shared quota; retries with provider-suggested backoff absorb it. Production would use per-user keys.

3. What I'd do differently with more time

- **A real fiscal calendar.** Quarters are currently calendar quarters, but the invoice numbering (`SDPL/FY25-26/916`) shows an April–March fiscal year — so quarter questions are probably being answered on the wrong boundaries.
- **Reconcile the two value sources.** Deal value is recorded on only ~40% of closed deals, yet work orders carry full amounts. Won deal values likely live on the Work Orders board; joining them would lift historical coverage from 40% to near-complete.
- **Fuzzy matching on the join.** Name matching is exact after lowercasing; the ~10% unmatched work orders may simply be spelling variants.
- **Trend analysis.** Every metric is a snapshot. Pipeline movement, stage velocity and month-on-month billing all need history the current fetch doesn't retain.
- **Tests and evaluation.** No test suite for the analysis functions, and no regression set of questions with expected tool calls. Both cut for time.
- **Write-back.** Read-only was specified, but the highest-value follow-up is flagging deals with missing values directly on the board — the agent can already identify every gap it complains about.

4. How I interpreted "prepare data for leadership updates"

- A leadership update is not a dashboard dump. It is the small set of things a founder would be embarrassed not to know before a board meeting — **including what the data cannot tell them.**
- `leadership_brief()` returns one bundle: pipeline by stage and probability, revenue and collections, delivery risk, sector performance, win rate, the top five open deals by value, live coverage figures, and an explicit `known_blind_spots` list.
- That last field is the substance. An update stating ₹10.7 crore billed without noting it covers 64% of work orders is worse than no update, because it invites confident action on an incomplete number. Every figure carries the share of records behind it, and the interface gives coverage its most distinctive visual treatment rather than a footnote.
- **This shaped the analysis, not just the presentation.** Asked about win rate, the totals suggest lost deals are 19× the size of won ones. The medians say 2.5×. The real signal is the maximum: largest deal ever won ₹1.46 crore, largest lost ₹75.1 crore — an apparent deal-size ceiling. One column, three different answers. The agent therefore reports distributions and flags when a mean is driven by outliers, rather than returning the first number it finds.

5. Data quality found in the source boards:Surfaced at runtime rather than cleaned away, since handling mess was the brief:

Defect	Handling
Two header rows embedded inside the Deals data	Detected structurally by value-equals-column-name, dropped
Literal `#VALUE!` Excel error in a money column	Parsed to null, counted
Month names, no year, inconsistently abbreviated (`Dec` beside `November`)	Mapped to month numbers; year never inferred
Casing typo `BIlled` / `Billed`; free text `Billed- Visit 7`	Canonicalised
Four columns 100% empty (collection dates and status)	Reported as untracked, never as zero
GST basis mismatch — collections incl. GST, billings excl.	Compared on a consistent basis: 84.2% → **71.4%**
Recorded receivable vs billed-minus-collected	Reconciles exactly once GST bases match
Sector concentration	Flagged where one deal exceeds 50% of a sector's value (Powerline: 92%)

6. Resilience notes

During the build the model provider **retired two model versions** and returned **503 overload** on a third. Model resolution therefore reads the replacement name out of the provider's own error response rather than a static list, retries quota limits using the server's suggested delay, and falls back across models on sustained overload.Pagination follows monday's cursor and is verified against each board's reported `items_count`. A single-page fetch would have silently analysed 100 of 344 deals and reported confident, wrong totals — the failure mode most likely to go unnoticed.If one board fails, the other still loads and questions about it are answered rather than the whole app failing.